

CS4234: Optimization Algorithms
Semester 1, AY 2026-27
School of Computing, NUS

Problem Set for Tutorial 2

Question 1: Traveling Salesperson Algorithms

Let $G = (V, E)$ be a complete weighted graph on n vertices, with edge cost $c(u, v)$. A traveling salesperson tour is a minimum-cost Hamiltonian cycle.

1A. An exact exponential-time algorithm

Give a dynamic-programming algorithm for the Traveling Salesperson Problem that runs in $O(2^n n^2)$ time. Clearly define the subproblems, recurrence, base cases, and final answer. You may fix an arbitrary vertex r as the start and end of the tour.

1B. The metric TSP approximation need not be optimal

Recall the double-tree 2-approximation for metric TSP: compute a minimum spanning tree, double its edges, find an Eulerian tour, and shortcut repeated vertices. Give a metric instance and a valid execution of this algorithm for which the returned tour is not optimal. State the costs of the returned tour and of a strictly better tour.

Question 2: Building Roads for Eulerian Tours

Humperdink and Mildred run two hotels in a popular tourist destination. A tourist wants to leave a hotel, traverse every road exactly once, and finish at the Central Plaza. In the current road network, neither hotel's guests can do so.

The network is a multigraph, so a newly built road may run parallel to an existing road.

2A. A route for Hotel Humperdink

Humperdink will build one new road. Which two locations should it connect so that a guest can start at Hotel Humperdink, traverse every road exactly once, and finish at the Central Plaza? Draw the new road on Figure 1 and justify your answer.

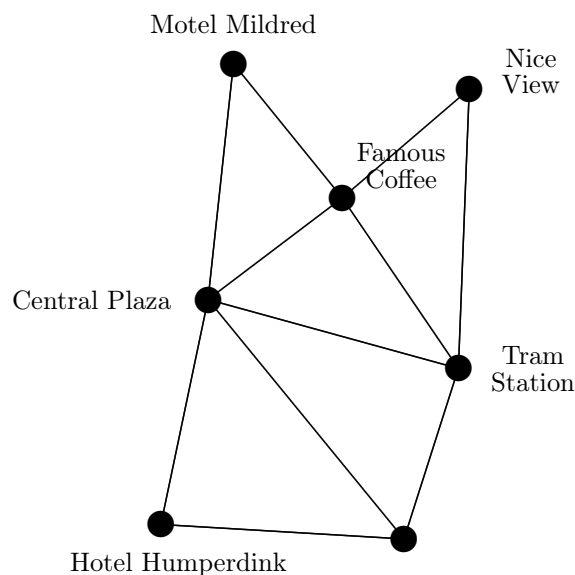


Figure 1: The town's road network. We call the unlabelled vertex in the lower-right corner the *South Junction*.

2B. A route for Motel Mildred, but not Hotel Humperdink

Starting from the network after Part 2A, Mildred will build one additional road. Which two locations should it connect so that a guest can start at Motel Mildred, traverse every road exactly once, and finish at the Central Plaza, while no such route starts at Hotel Humperdink? Justify your answer.

2C. Characterizing Eulerian paths

Let G be a connected multigraph, and let s and d be two distinct designated vertices. An *Eulerian path from s to d* starts at s , ends at d , and traverses every edge exactly once. Prove that G has such a path if and only if s and d have odd degree and every other vertex has even degree.

Question 3: Group Admission at a Museum

A museum can admit at most n visitors in one day. There are m groups, and group i contains g_i visitors. A group must be admitted in full or rejected in full. The objective is to maximize the total number of admitted visitors. Assume that g_i and n are positive integers.

3A. Analyzing a greedy algorithm

Consider the following algorithm. It processes the groups in nonincreasing order of size and admits a group whenever it fits in the remaining capacity.

Algorithm 1: Largest-first group admission

```
1 Function GroupAdmission( $g[1..m], n$ )
2   reorder the groups so that  $g_1 \geq g_2 \geq \dots \geq g_m$ 
3    $admitted \leftarrow 0$ 
4    $remaining \leftarrow n$ 
5   for  $i \leftarrow 1$  to  $m$  do
6     if  $g_i \leq remaining$  then
7       ADMIT( $i$ )
8        $remaining \leftarrow remaining - g_i$ 
9        $admitted \leftarrow admitted + g_i$ 
10    else
11      REJECT( $i$ )
12  return  $admitted$ 
```

Prove that this is a 2-approximation: if an optimal solution admits OPT visitors, the algorithm admits at least $OPT/2$ visitors.

3B. Showing that the analysis is tight

Give a family of instances for which the ratio ALG/OPT approaches $1/2$ as the capacity n grows.

3C. A read-only, constant-space algorithm

Now suppose that the array $g[1..m]$ is stored in read-only memory. You may scan it repeatedly, but you may not reorder it, modify it, or copy it, and you may use only $O(1)$ extra space.

Give a polynomial-time 2-approximation under these restrictions. Aim for an $O(m)$ -time algorithm. *Hint: First find the largest group that fits by itself, and consider separately whether its size is at least $n/2$.*